

• Machine Learning Engineer & AI Researcher with **7+ years** of experience in reinforcement learning, large language models, and scalable ML systems. **Awarded Best MSc Thesis** (CAIAC 2024) for advancing model-based reinforcement learning. Previously built AI-driven ranking and search systems serving 40M+ users. Published in ICML, NeurIPS, and CoLLAs, with expertise in deploying ML at scale. Passionate about bridging research and real-world applications, from LLM alignment to decision-making AI.

Experience

- **Mila - Quebec AI Institute** — *ML Research Assistant* Montreal, Canada — Dec 2023 – Present
 - Conducting research on AI alignment and interpretability by steering LLMs through activation space manipulation.
 - Exploring scalable deep reinforcement learning architectures and world modeling using large foundation models for model-based RL.
- **Mila - Quebec AI Institute** — *MSc Research Student* Montreal, Canada — Jan 2021 – Dec 2023
 - Researched model-based reinforcement learning and various forms of adaptation in continual learning, publishing in top AI conferences.
- **iCAD Dental** — *ML Research Developer (part-time)* Ottawa, Canada — Jan. 2022 – May. 2022
 - Designed and implemented a Python pipeline for classification and clustering of dental crown images, improving automated dental analysis.
- **CafeBazaar Inc.** — *Machine Learning Engineer* Tehran, Iran — Sep. 2017 – Jan. 2021
 - CafeBazaar is Iran’s largest third-party mobile app marketplace, **serving 40M+ users**.
 - Personalization Team:
 - Developed and deployed an app recommender system, improving **install rates from 1.6% to 2.5%**.
 - Designed a diversification strategy to broaden recommended app categories, enhancing user engagement.
 - Search-ads Team:
 - As a core founding member, led data science, mechanism design, and optimization for the search advertising system. ([Technical Report](#))
 - Helped establish Search-Ads as a top 3 daily revenue driver for the company.
 - Search Team:
 - Improved the app ranking system and built NLP-based services like spell-checking and query suggestion.
 - Reduced the **search bounce rate from 41% to 35%**, increasing search satisfaction.
- **Chair for CAMPAR** — *Research Assistant* Munich, Germany — Jul. 2019 – Oct. 2019
 - Worked on 3D camera re-localization in dynamic environments. Developed a valid pose generator using rendered scenes for 3RScan data.

Education

- **University of Montreal - Mila**
M.Sc. in Computer Science (Artificial Intelligence)
- **Sharif University of Technology**
B.Sc. in Computer Science

Publications

* DENOTES EQUAL CONTRIBUTION.

- **Steering Large Language Model Activations in Sparse Spaces**
Under Review ([paper](#))
Rahimi-Kalahroudi, A.*, Bayat, R.*, Pezeshki, M., Chandar, S., Vincent, P.
- **Partial Models for Building Adaptive Model-Based Reinforcement Learning Agents**
Conference on Lifelong Learning Agents, **CoLLAs 2024** ([paper](#))
Alver, S., **Rahimi-Kalahroudi, A.**, Precup, D.
- **Replay Buffer with Local Forgetting for Adapting to Local Environment Changes in Deep MBRL**
Conference on Lifelong Learning Agents, **CoLLAs 2023 (Oral)** ([paper](#))
and **NeurIPS 2022** - Deep Reinforcement Learning Workshop
Rahimi-Kalahroudi, A., Rajendran, J., Momennejad, I., Van Seijen, H., Chandar, S.
- **Towards Evaluating Adaptivity of Model-Based Reinforcement Learning Methods**
International Conference on Machine Learning, **ICML 2022** ([paper](#), [code](#))
Rahimi-Kalahroudi, A.*, Wan, Y.*, Rajendran, J., Momennejad, I., Chandar, S., Van Seijen, H.

Honors and Awards

- **Best M.Sc. Thesis** - Canadian Artificial Intelligence Association ([CAIAC](#)) Guelph, Canada — 2024
- Exceptional Grade for the M.Sc. Thesis (Top 1%) - University of Montreal Montreal, Canada — 2023
- Bourse D’excellence DIRO - University of Montreal Montreal, Canada — 2022
- M.Sc. Tuition Exemption Scholarships (Bourse C) - University of Montreal Montreal, Canada — 2021
- [International Undergraduate Excellence Award](#) - Technical University of Munich Munich, Germany — 2019
- **Silver Medal** - 24th Iranian National Olympiad in Informatics ([INOI](#)) Tehran, Iran — 2014

Skills

- **Programming:** Python, Scala, and C++.
- **ML Tools:** PyTorch, Tensorflow, NumPy, and JAX.
- **Data Technologies:** Apache Spark, Hadoop, Airflow, and Kafka.
- **Misc:** Docker, Apptainer, Bash, Slurm, and L^AT_EX